



Cambridge IGCSE™

PHYSICS

0625/61

Paper 6 Alternative to Practical

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MARK SCHEME

Maximum Mark: 40

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **14** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): - The response should be read as continuous prose, even when numbered answer spaces are provided. - Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. - Incorrect responses should not be awarded credit but will still count towards <i>n</i>. - Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. - Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	arithmetic error
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	response has not answered question

Annotation	Meaning
RE	rounding error
SEEN or 	point has been noted, but no credit has been given or blank page seen
SF	error in number of significant figures
TE	transcription error
TV	response is too vague or there is insufficient detail in response
T	answer outside the tolerance of the mark scheme
	used to highlight parts of an extended response
	used to highlight parts of an extended response
MO	mandatory mark not awarded
SC	special case
	unclear response
PD	poor diagram
POT	power of ten error
XP	incorrect physics
U	incorrect unit

Acronym / shorthand	Explanation
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word/phrase/unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
ORA	Or reverse argument
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, <u>not</u> treated as contradictory.
cao	correct answer only

Question	Answer	Marks	Guidance												
1(a)	Graph: Table 1.1 <table><tr><td>x / cm</td><td>y / cm</td></tr><tr><td>40.0</td><td>27.0</td></tr><tr><td>35.0</td><td>24.5</td></tr><tr><td>30.0</td><td>19.5</td></tr><tr><td>25.0</td><td>17.0</td></tr><tr><td>20.0</td><td>13.5</td></tr></table>	x / cm	y / cm	40.0	27.0	35.0	24.5	30.0	19.5	25.0	17.0	20.0	13.5		
	x / cm	y / cm													
	40.0	27.0													
	35.0	24.5													
	30.0	19.5													
25.0	17.0														
20.0	13.5														
Axes correctly labelled with quantity and unit the right way round		1	If wrong graph is plotted (e.g. x against y), this mark is not gained but other 3 are available ECF.												
Suitable scales		1	Do not award if origin present / reverse scales / awkward scales e.g. 3's, 7's etc. ignore (0, 0) if numbers on axes suggest a false origin required to score MP2 plots to occupy more than ½ the grid in both directions												
All plots correct to ½ small square		1	Indicate incorrectly plotted point e.g. highlight tool Do not allow large 'blobs' bigger than half a square												
Good line judgement, thin, continuous line		1	based on all the plots; not dot-to-dot; allow straight line between 1st and last plot for correctly plotted points												

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Question	Answer	Marks	Guidance
1(b)	Triangle method clearly shown on graph	1	allow two points indicated on line of best-fit
	Using at least half of distance between extreme plots	1	This point dependent on MP1 being given
	G in range 0.60 to < 0.75 seen as a decimal	1	allow G to any number of significant figures ignore candidate's calculation ignore fractions and units No ECF for G for x plotted against y ignore negative sign if G is negative
1(c)	Correct method seen	1	See ($W_Q =$) $2(.0) / \text{their } G$ or their $G = 2(.0) / W_Q$
	W_Q present with unit N	1	Ignore any working beyond correct method do not accept lower case n
1(d)	mark centre of (side of) P and align with 10.0 cm mark on ruler OR use scale readings either side (and average to give 10.0)	1	context; addressing difficulty of marks on ruler being covered, Fig. 1.1 has numbers (0 and 100) on side of ruler and line at 10 cm marked. allow object suspended from ruler with a thread allow use centre of slot in slotted mass ignore descriptions of parallax
	Diagram correctly illustrating method	1	Award this mark if there is no explanation but a correct diagram with the centre line labelled and 10 cm mark clear OR award this mark for the scale readings either side method if there is no explanation but a well-labelled diagram shows this clearly.

Question	Answer	Marks	Guidance
2(a)(i)	$I_1 = 0.44$	1	cao
2(a)(ii)	$V_1 = 1.8(0)$	1	cao
	units A and V	1	Allow units in words
2(a)(iii)	$R_1 = 4.1$	1	allow 4.09 (recurring) ECF to 2 or more significant figures allowed
2(a)(iv)	r correct	1	expect 0.082 ECF to 2 or more significant figures allowed If R_1 given as 4 accept $r=0.08$ Allow 0.0818 (recurring)
2(b)(i)	$N = 7$	1	cao not 7.0
2(b)(ii)	Correct calculation of l_1	1	Accept any answer that rounds to 51 ECF allowed for wrong N if correct to 2 significant figures NOT in terms of π e.g. 16.1π
2(c)(i)	$R_2 = 4.4$	1	Allow two or more significant figures (4.3902439) Not 4.40
2(c)(ii)	l_2 calculated and to nearest mm	1	Any value

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Question	Answer	Marks	Guidance
2(d)	Suitable reason (that matches suggestion)	1	no mark just for the suggestion (which could be l_1 or l_2) Example 1: l_2 (more accurate) because whole number of turns underestimates / gives inaccurate length owtte (MP1). Electrical measurements are more precise / accurate/reliable OR electrical method uses whole length of the wire (MP2) Example 2: l_2 (more accurate) because using a ruler to measure the (rod) diameter is inaccurate owtte (MP1) Electrical measurements are more precise / accurate/reliable (MP2) Example 3: l_1 (more accurate) because fewer readings required owtte (MP1) electrical method involves more measurements (MP2) <u>OR</u> other valid reasons ignore poor experimental practice e.g. parallax / human error
	Sensible comparison with other method	1	

Question	Answer	Marks	Guidance
3(a)	1.4	1	Cao not 1.40
	cm	1	Accept mm if h_0 in millimetres
3(b)(i)	$v = 9.3$	1	If a DFD script: If out of range (expect lower), if present check v with on screen ruler
3(b)(ii)	$V = 5 v$ expect 47	1	ECF allowed, $9.3 \times 5 = 46.5$
3(c)	Correct calculation of m_1 2.4	1	ECF allowed ($3.3 / 1.4 = 2.35714$) accept any number of significant figures ignore multiplication sign x ignore units
3(d)	Correct calculation of m_2 0.4(3)	1	ECF allowed ($0.6 / 1.4 = 0.42857$) ignore multiplication sign x ignore units
	m_2 to 1 significant figure	1	accept 2 significant figures
3(e)	Statement matches results	1	Expect yes OR $m_1.m_2$ should be 1.00
	Justification matches correct statement	1	Value is (very) close to 1.00 / within 10% of 1.00 / within the limits of experimental accuracy Check justification if $m_1.m_2$ not seen Statement NO if difference > 10% Value is much bigger/smaller than 1.00 / beyond 10% of 1.00 / beyond the limits of experimental accuracy ECF allowed Ignore 'my product rounds to 1.00' Award BOD to correct justification if no statement

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Question	Answer	Marks	Guidance
3(f)	Darkened room / bright object	1	
	Move screen slowly OR move screen back and forth OR approach clearest position of screen from both directions (and average)	1	Do not allow moving lens or object Accept 'it' for screen

Question	Answer	Marks	Guidance
4	MP1 ADDITIONAL APPARATUS: timer / stop-watch / clock	1	may be seen in a diagram, check page 10
	MP2 METHOD MARK 1: Heat water in one container (on the hotplate). Measure OR see OR record time taken for water to reach boiling point.	1	accept reference to (start) bubbling for boiling accept start AND stop timer for measuring time not calculate or 'investigate time taken to boil water' Ignore timing intermediate temperatures
	MP3 METHOD MARK 2: Repeat with a different container	1	Stated; not just from table.
	MP4 and MP5 TWO VARIABLES TO KEEP CONSTANT: Any two from: Volume of water Starting temperature of water / container Setting of hotplate	2	allow amount / quantity / mass of water
	MP6 TABLE with clear columns for container/metal (A, B, C, D, and E) with no unit and time with unit.	1	Allow h, min, s; not sec or secs Allow headed rows instead of columns Ignore any additional columns
	MP7 CONCLUSION: Draw / construct / plot a graph / chart with container / metal on one axis and time on the other axis. OR Compare times or rates for each container to see if / how / whether the container affects the time (to reach boiling point) OR see if the differences in times taken are significant / large / beyond experimental accuracy owtte	1	Accept a sketch with labelled axes Ignore the container which reaches boiling point in shortest time is the best owtte Ignore any reference to colour of containers Ignore a prediction